

GridOps Intelligence

Energy Data Engineering, Forecasting, and MLOps Platform

A production-style system that turns Ontario electricity and weather data into day-ahead forecasts, quality signals, operational alerts, and planning scenarios.

Portfolio and career development plan

Master project plan | Version 1.0 | July 2026

Designed for milestone-based development using ChatGPT, Codex, GitHub, CI, and repository handoffs.

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1. Executive summary and positioning

GridOps Intelligence is an end-to-end energy data platform focused on Ontario electricity-demand operations. It will ingest public energy and weather data, preserve raw source history, validate and transform the data, forecast hourly demand for the next day, quantify uncertainty, detect attention conditions, run planning scenarios, and expose results through an API and operational dashboard.

Primary user	Utility operations, planning, forecasting, analytics, and grid-modernization professionals.
Core question	What hours in the next 24 hours require attention, why are they unusual, and how confident is the forecast?
Portfolio identity	Energy data engineer and applied ML builder focused on operational decision systems.
Project balance	Approximately 65 percent data engineering and 35 percent machine learning.
Repository	gridops-intelligence
Delivery model	One ChatGPT thread per milestone, one Codex task and branch per ticket, and a verified repository handoff after every milestone.

2. Industry problem and product scope

Problem

Energy analysts work with demand observations, weather forecasts, calendar effects, generation context, data revisions, forecast performance, and source-health information that may arrive through separate feeds and reports. The operational challenge is not only producing a forecast. It is determining which periods need attention, whether the inputs are trustworthy, what is driving the forecast, and how planning assumptions change the expected demand profile.

Included in version 1

- Historical and incremental demand ingestion.
- Immutable raw-source snapshots with publication and retrieval metadata.
- Weather observations and archived weather forecasts.
- Day-ahead hourly Ontario demand forecasting.
- Peak-demand and peak-hour prediction.
- Prediction intervals and forecast-confidence indicators.
- Forecast-versus-actual monitoring.
- Data-quality, source-freshness, ramp, deviation, and high-demand alerts.
- Scenario analysis for weather and demand-growth assumptions.
- Deterministic daily analyst briefing.
- FastAPI services, dashboard, CI, deployment, and model monitoring.

Explicitly excluded

- Real electricity-grid dispatch or automated control.
- Emergency or reliability declarations.
- Trading recommendations.
- Power-flow calculations.
- Claims that the product replaces official system-operator forecasts.
- Unsupported causal claims or invented performance metrics.

3. Users, workflows, and measurable outcomes

User	Current pain	GridOps support
Forecasting analyst	Combines demand, weather, and calendar data manually.	Validated feature pipeline, backtesting, forecast intervals, and driver explanations.
Operations analyst	Must identify unusual hours and large deviations quickly.	Attention alerts, ramp detection, peak forecast, and daily briefing.
Data engineer	Manages changing feeds, revisions, missing intervals, and duplicate loads.	Contracts, immutable raw storage, idempotent pipelines, lineage, and observability.
Planning analyst	Needs sensitivity analysis for heat, load growth, or new large customers.	Controlled scenario engine with clearly labelled assumptions.
Model reviewer	Needs evidence that the model is leakage-safe and stable.	Time-based validation, baseline comparisons, slice metrics, registry, and drift monitoring.

Primary success metrics

- Mean absolute error, root mean squared error, weighted absolute percentage error, and forecast bias.
- Peak-demand error and peak-hour error.
- Prediction-interval coverage and width.
- Maximum-ramp forecast error.
- Pipeline success rate, freshness, completeness, duplicate rate, and recovery time.
- Alert precision, false-alert rate, and analyst usefulness during manual validation.
- Runtime, reproducibility, and degraded-input reliability.

4. Data strategy and engineering design

Core data sources

Source category	Purpose	Examples
IESO demand	Historical training data and current actuals.	Hourly Ontario demand, five-minute demand, projected demand.
IESO forecasts	External operational context and forecast comparison.	Predispatch totals, published demand forecasts.
Variable generation	Renewable context and future net-load experiments.	Wind and solar forecast summaries.
Weather observations	Ground-truth weather and feature validation.	Temperature, humidity, wind, precipitation, degree values.
Archived weather forecasts	Leakage-safe historical day-ahead model inputs.	Forecasts selected by issue time and lead time.
Calendar	Repeated demand patterns and special-day effects.	Hour, weekday, season, holidays, daylight-saving transitions.

Bronze, silver, and gold layers

Bronze	Original files or payloads, file hashes, source URLs, publication times, retrieval times, parser versions, and ingestion-run identifiers.
Silver	Normalized UTC and Toronto timestamps, source-native hour-ending fields, standard units, revision metadata, and quality flags.
Gold	Hourly demand facts, weather features, training snapshots, predictions, evaluation metrics, alerts, scenario results, and briefing facts.

Engineering requirements

- Historical backfills and scheduled incremental loads.
- Idempotent processing: rerunning a successful job must not create duplicate facts.
- Safe retries, failed-run recovery, and clear run-level logs.
- Late-arriving data and source revision handling.
- Schema, range, completeness, uniqueness, freshness, and continuity checks.
- Explicit processing of 23-hour and 25-hour daylight-saving days.
- Data lineage from raw source through model output.
- dbt transformations and documentation where they improve clarity.
- Prefect orchestration for scheduled ingestion, backfills, training, and inference.

5. Forecasting and MLOps strategy

Primary target

Forecast Ontario hourly electricity demand for the next 24 hours using only information available at the defined forecast-issuance time.

Baselines and candidates

Tier	Models	Purpose
Baseline	Same hour yesterday, same hour last week, seasonal hourly mean.	Establish the minimum acceptable performance.
Interpretable	Ridge regression with demand lags, calendar, and weather.	Provide a transparent statistical benchmark.
Production candidate	LightGBM or XGBoost.	Capture nonlinear weather, calendar, and lag interactions.
Uncertainty	Quantile gradient boosting.	Generate P10, P50, and P90 intervals.
Statistical comparison	SARIMAX.	Provide an independent time-series reference.

Core features

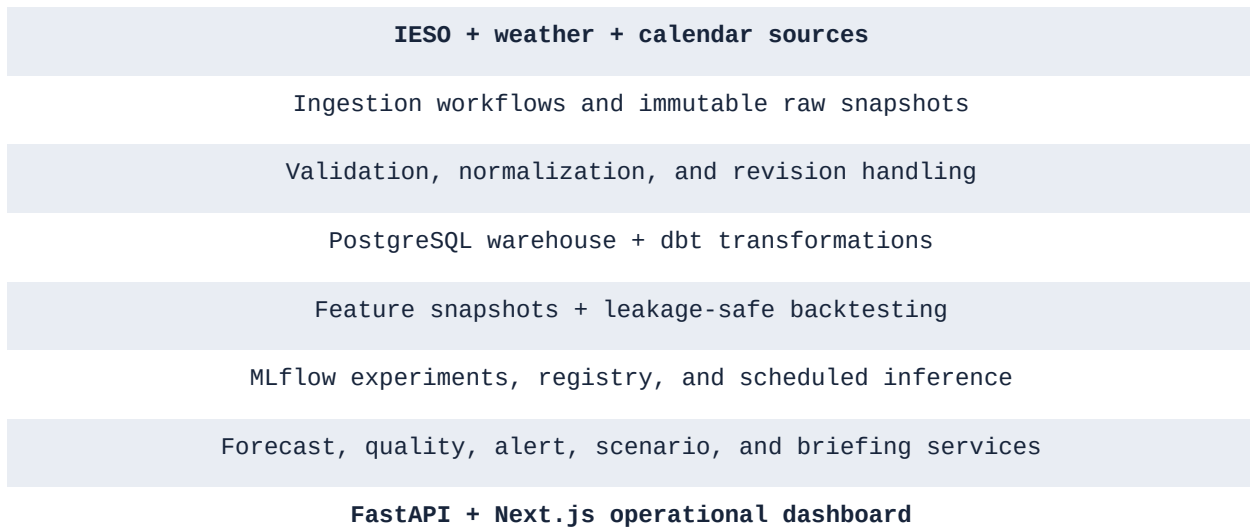
- Demand lags at 1, 2, 24, 48, and 168 hours.
- Rolling means, variability, and recent ramp features.
- Hour, weekday, month, season, holidays, and cyclic encodings.
- Regional forecast temperature, humidity, apparent temperature, wind, and degree values.
- Extreme-weather duration and temperature-hour interactions.
- Optional wind and solar forecast context.
- Previous forecast error and source-freshness indicators.

Non-negotiable ML rules

- Use rolling or expanding time-based backtesting, never random splitting.
- Join archived forecasts using publication-time as-of logic.
- Keep a final test period untouched during model development.
- Select the simplest model that meets overall and slice-level performance gates.
- Track data version, feature version, training window, parameters, metrics, code commit, and model artifact.
- Monitor forecast error, bias, interval calibration, drift, and degraded-input behavior.

6. Architecture and technology stack

Architecture overview



Recommended stack

Area	Technologies
Language and API	Python 3.12, FastAPI, Pydantic, HTTPX.
Database	PostgreSQL, SQLAlchemy, Alembic.
Pipelines	Prefect, dbt, Pandera or Great Expectations.
ML and MLOps	scikit-learn, LightGBM or XGBoost, statsmodels, SHAP, MLflow.
Frontend	Next.js, TypeScript, ECharts or Recharts.
Infrastructure	Docker Compose, GitHub Actions, object storage, managed PostgreSQL, container deployment.
Testing	Pytest, integration tests, API contract tests, migration tests, frontend smoke tests.

7. Milestone roadmap

M01 - Domain contract and foundation

Objective: Create a reproducible repository, API and database foundation with explicit time and quality rules.

Core deliverables

- Python project, formatting, linting, typing, tests, and CI.
- FastAPI health and readiness endpoints.
- PostgreSQL, SQLAlchemy, Alembic, Docker, configuration, and logging.
- UTC, Toronto time, DST, and IESO hour-ending contract.

Completion gate: Repository installs and runs reproducibly; CI passes; time rules are documented and tested.

M02 - Data ingestion and normalized storage

Objective: Acquire, archive, normalize, and track the core energy and weather sources.

Core deliverables

- Raw snapshot storage and source metadata.
- Historical and incremental IESO demand ingestion.

- Weather observation and archived forecast ingestion.
- Revision handling and ingestion observability.

Completion gate: Representative backfills and incremental runs are idempotent, traceable, and recoverable.

M03 - Data quality and observability

Objective: Prevent bad or stale source data from contaminating gold tables or model training.

Core deliverables

- Data contracts and severity levels.
- Completeness, continuity, uniqueness, range, schema, and freshness checks.
- DST validation and quality-result storage.
- Source-health API or operational report.

Completion gate: Defined failures are detected, persisted, tested, and capable of blocking downstream publication.

M04 - Baselines and backtesting

Objective: Create trusted feature snapshots and reproducible leakage-safe model evaluation.

Core deliverables

- Forecast-issue contract and as-of joins.
- Gold feature tables.
- Seasonal-naïve and ridge baselines.
- Rolling evaluation and slice reports.

Completion gate: Every training row is point-in-time safe; metrics and lineage are reproducible.

M05 - Production forecasting and MLOps

Objective: Train, select, register, schedule, serve, and monitor the production model.

Core deliverables

- Gradient-boosting and quantile models.
- Peak and ramp outputs.
- MLflow registry and model-selection gate.
- Scheduled inference, forecast API, drift, and performance monitoring.

Completion gate: Production candidate beats the agreed baseline, passes slice gates, and runs reproducibly.

M06 - Alerts, scenarios, and briefings

Objective: Turn forecast and quality outputs into useful operational attention signals.

Core deliverables

- Deviation, ramp, high-demand, and combined context alerts.
- Alert lifecycle and resolution tracking.
- Scenario engine for weather and load assumptions.
- Deterministic briefing and guarded optional narrative generation.

Completion gate: Every alert and briefing fact has deterministic evidence; scenarios are clearly labelled.

M07 - Dashboard, deployment, and release

Objective: Deliver the complete public-facing product and portfolio evidence.

Core deliverables

- Responsive dashboard and API client.
- Forecast, alert, quality, scenario, and model-performance screens.
- Deployment, public demo mode, documentation, case study, and demo video.
- Full release verification and v1.0.0 tag.

Completion gate: Deployed workflows operate safely; all claims are backed by verified artifacts.

8. Testing, validation, and quality gates

- Parser and transformation unit tests.
- Database and migration integration tests.
- API schema and behavior tests.
- Ingestion idempotency, retry, and revision tests.
- Missing interval, duplicate, stale source, malformed schema, and out-of-range tests.
- DST transition and hour-ending tests.
- Point-in-time feature and leakage tests.
- Forecast reproducibility and model-promotion tests.
- Scenario and alert-threshold tests.
- Frontend smoke and end-to-end forecast-run tests.

Ticket completion gate

1. Acceptance criteria are satisfied.
2. Relevant checks pass and exact results are recorded.
3. The final diff contains no unexplained scope expansion.
4. Manual verification is reproducible.
5. CURRENT_STATE and affected specifications are updated.
6. Known limitations are stated honestly.

Milestone completion gate

7. All required tickets are merged.
8. The full supported test suite passes.
9. Manual milestone validation is completed.
10. Verification and handoff documents are finalized.
11. The roadmap and current state reflect repository reality.

9. Deployment and portfolio package

Deployment target

- Containerized FastAPI service.
- Managed PostgreSQL database.
- Object storage for raw snapshots and model artifacts.
- Scheduled orchestration for ingestion and inference.
- Next.js public dashboard.
- Read-only public mode with bounded temporary scenario execution.
- GitHub Actions for quality checks and deployment.

Portfolio deliverables

- Live demo and clean GitHub repository.
- Three-to-five-minute product video.
- Architecture and data-flow diagrams.
- Data dictionary and source contracts.
- Model card and backtesting report.
- Data-quality and observability report.
- API documentation and screenshots.
- Detailed case study with limitations and deployment notes.
- Verified v1.0.0 release.

10. Career alignment and final success criteria

Role family	Evidence produced by GridOps
Data engineer	Multi-source ingestion, idempotency, raw storage, warehouse design, dbt, orchestration, lineage, CI, and observability.
Analytics engineer	Gold tables, semantic definitions, data tests, dashboards, and business-facing metrics.
ML engineer	Feature pipelines, time-series backtesting, model registry, scheduled inference, API serving, drift, and reproducibility.
Energy data analyst	Demand, weather, peak, ramp, renewable context, and planning scenario analysis.
Applied AI engineer	Decision-support workflows, guarded narrative generation, explainability, testing, and production deployment.

Final success statement

GridOps succeeds when it is demonstrably more than a notebook: a tested, deployed, point-in-time-correct data and ML system that converts changing public energy data into reproducible forecasts and transparent operational decision support.

Reference data and documentation

IESO Data Directory - Official directory for Ontario power-market and demand reports.

<https://www.ieso.ca/power-data/data-directory>

IESO Real-time Demand Reports - Official description of real-time and predispatch demand reports.

<https://www.ieso.ca/power-data/demand-overview/real-time-demand-reports>

IESO Historical Demand - Official historical demand overview.

<https://www.ieso.ca/Power-Data/Demand-Overview/Historical-Demand>

IESO Public Reports - Official report directories and machine-readable files.

<https://reports-public.ieso.ca/public/>